Type MS · Finding a curve from its gradient function

Answer 1 of 6 · Question 7 part a.i · 4 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

7 a (i)	f $x = \int (4x^3 - 12x^2 - 19x + 12)\,dx = \frac{4}{4}x^4 - \frac{12}{3}x^3 - \frac{19}{2}x^2 + 12x + D$ oe For the point $4, -104$ it follows that $-104 = (4)^4 - 4(4)^3 - \frac{19}{2}(4)^2 + 12(4) + D$ * $-104 = 256 - 256 - 152 + 48 + D \Leftrightarrow D = 0$*	M1 A1 M1 A1* cso (4)
a (ii)	$x = 0.5$ f $'(x) = 4(0.5)^3 - 12(0.5)^2 - 19(0.5) + 12 = 0$ f$''(x) = 12x^2 - 24x - 19$ $x = 0.5$ f$''(x) = 12(0.5)^2 - 24(0.5) - 19 (= -28) < 0$ Therefore maximum *	M1 M1 A1 cso (3)
b (i)	f$'(x) = (2x-1)(2x^2 - 5x - 12)$ f$'(x) = (2x-1)(2x+3)(x-4) \rightarrow x =$ $x = -\frac{3}{2}$ or $x = 4$, $\left[x = \frac{1}{2}\right]$ $A\left(-\frac{3}{2}, -\frac{333}{16}\right)$ $B(4, -104)$	M1 M1 A1 A1 A1 (5)
b (ii)	f$''(x) = 12(-1.5)^2 - 24(-1.5) - 19 (= 44) > 0$ Therefore minimum $x = 4$ f$''(x) = 12(4)^2 - 24(4) - 19 (= 77) > 0$ Therefore minimum (Read notes carefully for allocation of these marks) **Alternative** As we have a maximum at $x = 0.5$ and C is a continuous curve $x = -\frac{3}{2}$ is a minimum and $x = 4$ is a minimum	ddM1 A1 A1 (3) {ddM1} {A1}{A1} (3)
		Total 15 marks

Type MS · Finding a curve from its gradient function

Answer 1 of 6 · Question 7 part a.i · 4 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

Part	Marks	Notes
(a) (i)	**M1**	Attempt to integrate – see general guidance for minimally acceptable attempt – at least one term must be fully correct in this integration (unsimplified).
	A1	Fully correct integration – does not need to include a constant of integration, need not be simplified.
	M1	Substitution of $x = 4$ and $y = -104$ into their expression, which must include a constant of integration.
	A1 cso	Obtains the given result, making it clear the constant of integration is 0 or listing fully the function at the end after fully correct working. There must be no incorrect working for this final mark to be awarded.
Note: candidates may assume $D = 0$ & show, by substituting, (4, -104) lies on the curve **final M1 A1**		
(a) (ii)	**M1**	Substitution of $x = 0.5$ into $f'(x)$ and shows $f'(x) = 0$. A candidate may also solve $f'(x) = 0$ and solve to get $x = 0.5$.
	M1	For finding $f''(x)$ - see general guidance for minimally acceptable attempt at differentiation. At least one term must be fully correct for this differentiation.
	A1 cso*	Substitutes $x = 0.5$ into a fully correct second derivative, shows that the result is negative and draws a conclusion. This can be as simple as "shown" or #. It is not necessary to work out the actual value of $f''(0.5)$ if clear substitution is shown, but if calculated, it must be correct. No incorrect work for this mark to be awarded. Note: as always, this A mark must follow 2 M1 marks.

Type MS · Finding a curve from its gradient function

Answer 1 of 6 · Question 7 part a.i · 4 marks

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(b)	Part (i) and (ii) may be marked together	
(i)	**M1**	A clear attempt to divide by $(2x-1)$ or compare coefficients. The student must arrive at a quadratic factor of the form $(2x^2+Ax-12)$ if dividing It is also possible to divide by $\left(x-\frac{1}{2}\right)$ and arrive at a quadratic factor of the form $4x^2+Bx-24$
	M1	Uses a complete method to solve their 3TQ, see general guidance for a minimally acceptable attempt. Must progress to $x=$.
	A1	$x=-\frac{3}{2}$ or $x=4$ Award M1 M1 A1 if **both** correct values are given without working.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ or $B(4,-104)$ Allow x = and y = (clearly paired) for **both** final A marks.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ and $B(4,-104)$ accept 20.8 or better – answer is 20.8125
	Factorising in (a)(ii) and then using in this part can be awarded the marks above.	
(b) (ii)	**ddM1**	Correctly substitutes $x=-\frac{3}{2}$ or $x=4$ into their $f''(x)$. Their values Dependent on both previous M marks in part (b). A fully correct evaluation of the second derivative can imply this mark ie sight of 77 and 44. Part (i) and (ii) may be marked together
	A1	Substitutes $x=-1.5$ **or** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion this is a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
	A1	Substitutes $x=-1.5$ **and** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion that both values of x give a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
ALT	**ddM1**	For stating the curve is continuous and there is a maximum at $x=0.5$ Dep both previous M
	A1	For stating "as we have a maximum at $x=0.5$" and progressing to state either A or B must be a minimum.
	A1	Fully correct conclusion, stating A and B must both be minimum points.
There **MUST** be an appropriate justification for these marks to be awarded, such as that given in M1		

Type MS · Finding a curve from its gradient function

Answer 1 of 6 · Question 7 part a.i · 4 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question number	Scheme	Marks
	4 ... 3×500 ...	M1 A1

Type MS · Finding a curve from its gradient function

Answer 2 of 6 · Question 9 part a · 4 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)\ \therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\quad \frac{d^2y}{dx^2}=12-4-4>0\quad \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\quad \frac{d^2y}{dx^2}=3-2-4<0\quad \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\quad \frac{d^2y}{dx^2}=12-4-4>0\quad \therefore$ min at $x=2$	A1cso
	$x=1\quad \frac{d^2y}{dx^2}=3-2-4<0\quad \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\quad \frac{d^2y}{dx^2}=12+4-4>0\quad \therefore$ min point	A1cso (3)
		[14]

Type MS · Finding a curve from its gradient function

Answer 2 of 6 · Question 9 part a · 4 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ ∴ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

Type MS · Finding a curve from its gradient function

Answer 2 of 6 · Question 9 part a · 4 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Finding a curve from its gradient function

Answer 3 of 6 · Question 8 part whole · 9 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part whole in this mark scheme extract

8	$\int\left(18x^2-2x+13\right)\,\mathrm{d}x=\frac{18x^3}{3}-\frac{2x^2}{2}+13x\left(+c\right)$	M1A1
	$\mathrm{f}\left(\frac{1}{2}\right)=6\times\left(\frac{1}{2}\right)^3-\left(\frac{1}{2}\right)^2+13\times\left(\frac{1}{2}\right)+c=0\Rightarrow c=-7$	M1A1
	$\mathrm{f}\left(x\right)=6x^3-x^2+13x-7$ oe	A1
	$2x-1\overline{)6x^3-x^2+13x-7}$ $3x^2+x+7$	M1A1
	$b^2-4ac=1^2-4\times3\times7=-83$	M1
	Conclusion: $b^2-4ac<0$ so there is only one root/intersection with the x-axis $\left[\text{at }\left(\frac{1}{2},0\right)\right]$	A1cso
		[9]
		Total 9 marks

Mark	Notes
M1	For integrating the given $\mathrm{f}'\left(x\right)$. $+c$ is not required for this mark This can be simplified or unsimplified. Accept $\frac{18x^3}{3}-\frac{2x^2}{2}+13x\left(+c\right)$ or $6x^3-x^2+13x\left(+c\right)$ At least two terms must be integrated correctly, with no differentiation.
A1	For the correct integrated expression. Ignore the absence of $+c$
M1	For substituting $x=\frac{1}{2}$ into their integrated expression (which must include $+c$, otherwise it is M0) and setting equal to 0. **ALT** Some candidates are completing this step using polynomial division. If they do not have $+c$ at this stage then it is M0. If it is clear they are attempting to find the value of c by finding a numerical value for c [even if it is incorrect], award both this mark and the next M mark provided they obtain a quotient of $3x^2+x+k$

Type MS · Finding a curve from its gradient function

Answer 3 of 6 · Question 8 part whole · 9 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

A1	For the correct value of $c = -7$
A1	For the fully correct $f(x) = 6x^3 - x^2 + 13x - 7$ all on one line. You do not need to see f(x) = … Accept even when you see this as a complete dividend in the 'bus stop' or when equating coefficients.
M1	Polynomial division For dividing their f (x) by $(2x - 1)$ to achieve as a minimum $3x^2 + x$. Award this for correct attempts even without either + c or a value for c. Allow also: $x - \frac{1}{2}\overline{)6x^3 - x^2 + 13x - 7}$ $\begin{matrix}6x^2 + 2x \ (+l)\end{matrix}$ [the correct quotient is $6x^2 + 2x + 14$] **ALT** Equating coefficients. They must achieve a correct value for A and B for the award of this mark. $6x^3 - x^2 + 13x - 7 = (2x-1)(Ax^2 + Bx + C) \Rightarrow A = 3, \quad B = 1$ OR $6x^3 - x^2 + 13x - 7 = \left(x - \frac{1}{2}\right)(Ax^2 + Bx + C) \Rightarrow A = 6, \quad B = 2$
A1	For the correct quotient/quadratic factor $3x^2 + x + 7$ or $6x^2 + 2x + 14$
M1	For using $b^2 - 4ac$ on their f (x) which must be a 3TQ Accept this as part of the quadratic formula. Sight of the complex roots $\left(\frac{-1 \pm (\sqrt{83})\mathrm{i}}{6}\right)$ is M1 Allow this mark even if they do not have +c or find the correct cubic expression.
A1 cso	For concluding that as $b^2 - 4ac < 0$ then there is only one root/intersection with the x-axis. Do not accept statements such as 'not possible' or 'will not factorise' without reference to a negative discriminant. Accept evidence of a statement such as $1^2 < 4 \times 3 \times 7 \Rightarrow b^2 < 4ac$ oe, or embedded in a formula without explicit evaluation. Complex roots must be followed by a comment that as the roots are not real, there are no intersections with the x-axis. Quoting the complex roots without a correct explanation is A0. If they have not found +c at the start, this mark is not available as this solution is cso.

Type MS · Finding a curve from its gradient function

Answer 3 of 6 · Question 8 part whole · 9 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding a curve from its gradient function

Answer 4 of 6 · Question 10 part a · 6 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

10 (a)	$\int\left(3x^2-4x-p\right)\,\mathrm{d}x=\frac{3x^3}{3}-\frac{4x^2}{2}-px+c\quad\left[=x^3-2x^2-px+c\right]$ $y=x^3-2x^2-px+c$ At $(2,0)\quad 0=8-8-2p+c\Rightarrow c=2p$ At $(-1,9)\quad 9=-1-2+p+c\Rightarrow c=12-p$ $\Rightarrow p=4, c=8\Rightarrow y=x^3-2x^2-4x+8$*	M1A1 M1 M1 A1A1cso [6]
(b)	$x^3-2x^2-4x+8=8-4x\Rightarrow x^3-2x^2=0\Rightarrow x^2(x-2)=0$ $x=0, x=2$ Area $=\int_0^2(8-4x)\,\mathrm{d}x-\int_0^2\left(x^3-2x^2-4x+8\right)\,\mathrm{d}x$ Area $=\int_0^2(8-4x)\,\mathrm{d}x-\int_0^2\left(x^3-2x^2-4x+8\right)\,\mathrm{d}x=\int_0^2\left(-x^3+2x^2\right)\,\mathrm{d}x$ Area $=\left[-\frac{x^4}{4}+\frac{2x^3}{3}\right]_0^2=\left(-4+\frac{16}{3}\right)-(0)=\frac{4}{3}$	M1 A1 M1 M1M1A1 [6]
		Total 12 marks

Type MS · Finding a curve from its gradient function

Answer 4 of 6 · Question 10 part a · 6 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding a curve from its gradient function

Answer 5 of 6 · Question 8 part a · 4 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

8(a)	$\int 17+2x-3x^2\ \mathrm{d}x = 17x+\frac{2x^2}{2}-\frac{3x^3}{3}+k$	M1A1
	$0=\left(17\times(-1)+\frac{2\times(-1)^2}{2}-\frac{3\times(-1)^3}{3}\right)+k \Rightarrow k=15$	M1
	$y=15+17x+x^2-x^3$	A1 [4]
(b)	$\frac{(15+17x+x^2-x^3)}{(x+1)}=-x^2+2x+15$	M1A1
	$-x^2+2x+15=(x+3)(5-x)$	M1A1
	$a=-3,\ [-1]$ and $b=5$	A1
	When $x=0$ $y=15$ so $c=15$	B1 [6]
(c)	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x-\frac{1}{2}\times5\times15$	M1
	OR	
	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x-\int_0^5(15-3x)\ \mathrm{d}x$	
	$\int_0^5(15+17x+x^2-x^3)\ \mathrm{d}x=\left[15x+\frac{17x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	$\left(15\times5+\frac{17\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)-(0)=\left(\frac{2075}{12}\right)$	M1
	Area of triangle Area $=\frac{1}{2}\times5\times15=37.5$ **OR**	B1
	Area $=\int_0^5(15-3x)\ \mathrm{d}x=\left[15x-\frac{3x^2}{2}\right]_0^5=37.5$	[B1]
	For the correct area of $R=\frac{2075}{12}-37\frac{1}{2}=\frac{1625}{12}=135\frac{5}{12}$	A1 [5]
		Total 15 marks

Type MS · Finding a curve from its gradient function

Answer 5 of 6 · Question 8 part a · 4 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Question	Notes	Marks
8(a)	$f'(x)=17+2x-3x^2$	
	For an attempt to integrate $f'(x)$ $y=17x+\frac{2x^2}{2}-\frac{3x^3}{3}+(k)$	M1
	For the correct integral **including** a constant of integration $y=17x+\frac{2x^2}{2}-\frac{3x^3}{3}+k$	A1
	For substituting $(-1\ ,\ 0)$ into their integrated expression, which must include a constant of integration. $0=\left(17\times(-1)+\frac{2\times(-1)^2}{2}-\frac{3\times(-1)^3}{3}\right)+k\Rightarrow(k=15)$	M1
	For writing the equation in the required form $y=15+17x+x^2-x^3*$ **This is a given equation. Every step above must be seen for the award of full marks.**	A1 cso [4]
(b)	Divides $(15+17x+x^2-x^3)$ by $(x+1)$ $x+1\overline{)15+17x+x^2-x^3}\quad Q+2x+x^2$ **OR** Equates coefficients $(15+17x+x^2-x^3)=(x+1)(Ax^2+Bx+c)\Rightarrow(x+1)(-x^2+2x+Q)$ Where Q is a constant **Minimal working here is sight of the quadratic factor.**	M1
	For obtaining the correct 3TQ $-x^2+2x+15$	A1
	For factorising their 3TQ [or otherwise solving] $-x^2+2x+15=(x+3)(5-x)$	M1A1
	For $a=-3$, $[-1]$ and $b=5$ identified clearly.	A1
	SC No working seen [use of a root finder on a calculator]	
	For $(x+1)(x-5)(x+3)$ seen leading to $a=-3$ and $b=5$ with no working award: M0A0MA1A1	
	For $a=-3$ and $b=5$ seen with no other working seen award: M0A0M0A0A1	
	For the value of $c=15$	B1
(c)	For writing a correct expression for the area of R with the correct limits $\int_0^5(15+17x+x^2-x^3)\,dx-\frac{1}{2}\times5\times15$ OR	M1

Type MS · Finding a curve from its gradient function

Answer 5 of 6 · Question 8 part a · 4 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	$\int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x - \int_0^5 (15-3x)\,\mathrm{d}x$	
	For an attempt to integrate the expression for the curve. [Ignore limits for this mark] Area $= \int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x = \left[15x+\frac{17x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	For evaluating their integral using their limits. Area $= \left(15\times5+\frac{17\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)-(0)=\left(\frac{2075}{12}\right)$	M1
	For the area of the triangle Area $=\frac{1}{2}\times5\times15=37.5$ **ALT** Integrates the line Area $= \int_0^5 (15-3x)\,\mathrm{d}x = \left[15x-\frac{3x^2}{2}\right]_0^5 = 37.5$	B1 [B1]
	For the correct area of R $\frac{2075}{12}-37\frac{1}{2}=\frac{1625}{12}=135\frac{5}{12}$	A1 [5]
	ALT	
	For writing an expression for the area of R with the correct limits. $\int_0^5 \left(15+17x+x^2-x^3\right)\,\mathrm{d}x - \int_0^5 (15-3x)\,\mathrm{d}x = \left[\int_0^5 \left(20x+x^2-x^3\right)\,\mathrm{d}x\right]$ Award the B mark for a correct expression for the combined area.	M1 B1
	For an attempt to integrate the expression for the combined area or just the curve. [Ignore limits for this mark] Area $= \int_0^5 \left(20x+x^2-x^3\right)\,\mathrm{d}x = \left[\frac{20x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}\right]_0^5$	M1
	For evaluating their integral using their limits, but the lower limit must be 0. Area $= \left(\frac{20\times5^2}{2}+\frac{5^3}{3}-\frac{5^4}{4}\right)=\left(\frac{1625}{12}\right)$	M1
	For the correct final area.	A1 [5]
		Total 15 marks

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2023 JAN · Paper 01 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding a curve from its gradient function

Answer 6 of 6 · Question 3 part a · 5 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

3(a)	$\left(y=\int\left(mx^2-10x-37\right)\mathrm{d}x=\right)\frac{mx^3}{3}-\frac{10x^2}{2}-37x\left(+c\right)$	M1
	$\left[\text{At } (1, 20)\right]$ $20=\frac{m\times1^3}{3}-\frac{10\times1^2}{2}-37\times1+c\left(\Rightarrow 62=\frac{m}{3}+c\right)$ oe	dM1
	$\left[\text{Factor of } (x-5)\right]$ $0=\frac{m\times5^3}{3}-\frac{10\times5^2}{2}-37\times5+c\left(\Rightarrow 310=\frac{125m}{3}+c\right)\Rightarrow m=6 \qquad c=60$	M1 M1
	$\Rightarrow\left(\mathrm{g}\left(x\right)=\right)2x^3-5x^2-37x+60$ *	A1cso [5]
ALT	3rd and 4th method marks $(x-5)\left(\frac{m}{3}x^2+bx\mp\frac{c}{5}\right)=\frac{m}{3}x^3-5x^2-37x\pm c\Rightarrow m=6 \qquad c=60 \qquad (b=5)$	M1 M1
Special case up to SC3 – mark as M1dM1M1 M0A0 in Epen	Differentiates the given expression for g (x)correctly to give $2(3x^2)-5(2x)-37\left(=6x^2-10x-37\right)$ **and** concludes $m = 6$	SC1
	Correctly substitutes $x = 5$ into the given function and shows g (x) = 0 $\left(\mathrm{g}(5)=\right)2\times5^3-5\times5^2-37\times5+60=0$	SC1
	Correctly substitutes $x = 1$ into the given function and shows and g (x) = 20 $\left(\mathrm{g}(1)=\right)2\times1^3-5\times1^2-37\times1+60=20$	SC1
(b)	$x-5\overline{)2x^3-5x^2-37x+60}$ $\quad 2x^2+5x-12$	M1
	$2x^2+5x-12=(2x-3)(x+4)$ $\left(\mathrm{g}(x)=(x-5)(2x-3)(x+4)=[0]\right)$	M1
	$\Rightarrow x=5,\ \frac{3}{2},\ -4$	A1 [3]
		Total 8 marks

Type MS · Finding a curve from its gradient function

Answer 6 of 6 · Question 3 part a · 5 marks

2023 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For a minimally acceptable attempt to integrate the given expression (see general guidance). No power of x to decrease. $+c$ is not required for this mark
	dM1	Correctly substitutes $x = 1$ and $y = 20$ to form an equation – simplification is not required Dependent on 1st method mark. A fully correct equation with no incorrect working can imply this mark. We will be lenient on not seeing the substitution of $x = 1$ as this is trivial, so long as the resulting equation is correct from their g (1) = 20. $+c$ is not required for this mark
	M1	Uses the information correctly that $(x – 5)$ is a factor of g (x) by correct substitution of $x = 5$ and $g(x) = 0$ into their g (x). Their g (x) **must** be a 4 term cubic expression. A fully correct equation with no incorrect working can imply this mark – but if the equation is incorrect we must see substitution of $x = 5$ such that $g(5) = 0$.
	M1	Uses a valid and complete method to solve their resulting simultaneous equations which must be in two variables. Allow up to 2 errors.
	A1 cso	For the correct function as shown or the correct expression. This is a show question. There must be no errors for the award of this mark.
ALT	3rd M1	Uses the information correctly that $(x – 5)$ is a factor of g (x) by writing a correct statement like the one shown for their g (x). Their g (x) **must** be a 4 term cubic expression.
	4th M1	Uses a valid and complete method to solve their resulting simultaneous equations in three variables. Allow up to 2 errors. It is not necessary to see the value for "b".
(b)	M1	For an attempt at polynomial division or equate coefficients to reach a 3 term quotient. The minimally acceptable quotient is $2x^2 \pm 5x \pm k \quad k \neq 0$ where k is a constant. In the absence of any other working, $2x^2 \pm 5x \pm k \quad k \neq 0$ may be accepted as evidence of working.
	M1	For attempting to factorise their quotient which must be of the form $2x^2 \pm 5x \pm k \quad k \neq 0$ - we must at least see a quadratic factor to factorise. See General Guidance for the definition of an attempt. It would also be possible to see a minimal attempt to use the quadratic formula or completing the square to solve (see general guidance) if their quadratic, of the form $2x^2 + 5x \pm k \quad k \neq 0$ does not factorise. If using the formula the substitution of their a, b and c must be completely correct.
	A1	For the correct values of x
Candidates have clearly been directed to use algebra in the question and therefore 0 marks can be scored without seeing M1 M1		

Type MS · Finding a curve from its gradient function

Answer 6 of 6 · Question 3 part a · 5 marks

2023 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks